Date: November 20, 2025

Biohybrid Compliant Mechanisms

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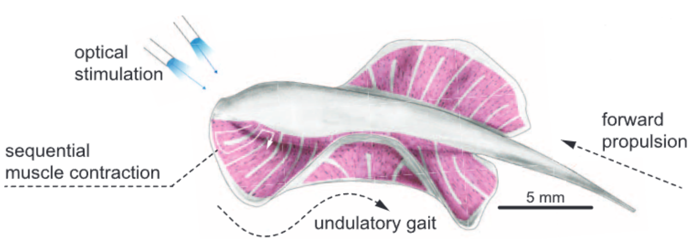
Introduction

Biohybrid machines involve synergistic integration of engineered abiotic with lab-grown muscles or other biotic components. Biological components have evolved over millennia to perform certain functions, some of which are not feasible to replicate with synthetic materials to the same level of functionality and performance. These features include self-replication, self-assembly, self-healing, and adaptive response. Additionally, they are an efficient mechanochemical engine, capable of deriving energy through their nutrient medium. On the other hand, synthetic engineering materials have allowed for highly performance-optimised mechanisms developed through a well-validated theoretical framework. Over the past two decades, there has been a growing interest in integrating both of their respective strengths. The field is still in its early stages, with most studies serving as proof-of-concept demonstrations rather than practical implementations. An example of such a system is illustrated in Fig. 1.

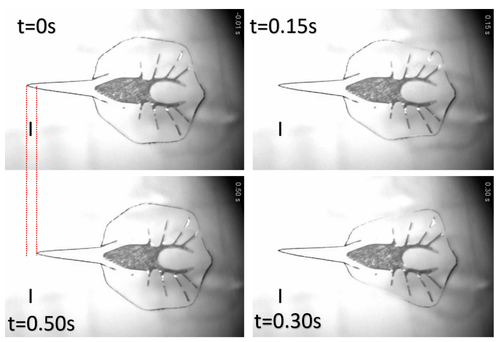
Tissue-engineered muscle cells are commonly employed in biohybrid machines as actuators due to their efficient actuation capabilities at low size scales (Huber et al. 1997). Their small length scale also allows for *distributed actuation* (as envisioned by Thurston (1895)), which is analogous to the concept of distributed compliance inherent in some compliant mechanisms. Currently, the use of large volumes of such actuators is limited due to the high cost of the extracellular matrix necessary for their growth, as well as the inability to provide vascular support networks to sustain life. Thus, with the limited actuation capacity, there is a need for compliant transmission systems to efficiently transmit the forces or motion. Motivated by these needs, we attempt to explore biohybrid machines from a structural perspective in our work. We mainly focus our attention on cell-actuated compliant mechanisms, defining terminology and developing effective interfaces between biotic and abiotic components.

Objectives

1. Design and fabricate cell-actuated compliant mechanisms. (preliminary work done)
2. Devise a strategy for finding the mobility (DoF) of biohybrid compliant mechanisms. (preliminary work done)



(a) Conceptual image



(b) Locomotion of the biohybrid fish

Figure 1: Biohybrid batoid fish (from Park et al. 2016)

3. A design method for low Reynolds number and high Reynolds number biohybrid swimmers. (future work)
4. Designing interfaces between biotic and abiotic components through topology optimisation. (future work)

Methodology

1. A design of a contact-aided compliant gripper with stiffness in the range of passive cell stiffness in the order of $\sim$1 mN/m is performed through non-dimensional kinetoelastic maps of compliant mechanisms from the M2D2 database[1]. Representative schematic is shown in Fig. 2b. Fabrication of these devices through bioprinting is currently underway.
2. In a manner similar to rigid-body manipulators, we sought to quantify the number of degrees of freedom of a biohybrid mechanism, with distributed compliance and actuation, such as that in Fig. 1 possesses. We have devised a strategy to count this number through spectral decomposition of the compliance matrix of the finite-element method. However, this produces deformation modes that may not be realisable in practice, as it does not account for the spatial distribution of actuators within the body. Modification to account for this has been performed through the construction of *actuation matrix* in the finite element framework.

   We have explored the degrees of freedom of a few biological specimens in a stingray and a seahorse. We also analysed the mobility of the biohybrid batoid fish shown in Fig. 1. Examples are shown in figs. 2c to 2j. Interestingly, we note that the eigendeformation modes were able to replicate lifelike motion. So far, we have only considered optimal deformation modes for an energy-like constraint (based on L_2 norm of forces) applied to distributed actuators, which will need to be modified in the future to a force-based constraint (based on L_1 norm of forces) for application to traditional compliant mechanisms, where actuation is typically concentrated.
3. With the deformation modes, it is possible to calculate the net translation and rotation of the system by suitably modelling the structure-environment interaction. In the future, we plan to calculate this motion for swimmers. In low Reynolds numbers, the linearity of the Stokes equation allows for geometric methods to represent gait as a response to the superposed deformation modes (Shapere et al. 1989). In high Reynolds number, the structure of Euler's equation allows for approximating flow by superposition of potential functions (irrotational flow) of different deformation modes, where the net translation in a time period is exact. We would like to utilise these ideas to conduct a highly idealised one-way fluid-structure interaction study to investigate the motion of biohybrid swimmers. These simple calculations enable us to modify locomotion patterns by linking them to the actuator distribution, which we aim to reposition to achieve our desired motion. An overview is presented in Fig. 2k.
4. In the future, we also want to study the design of compliant interfaces between biotic and abiotic components. We plan to focus on engineering the myotendinous junction (shown in Fig. 2l) for biohybrid systems through topology optimisation.

References

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Huber, JE et al. (1997). "The selection of mechanical actuators based on performance indices". In: *Proceedings of the Royal Society of London*.

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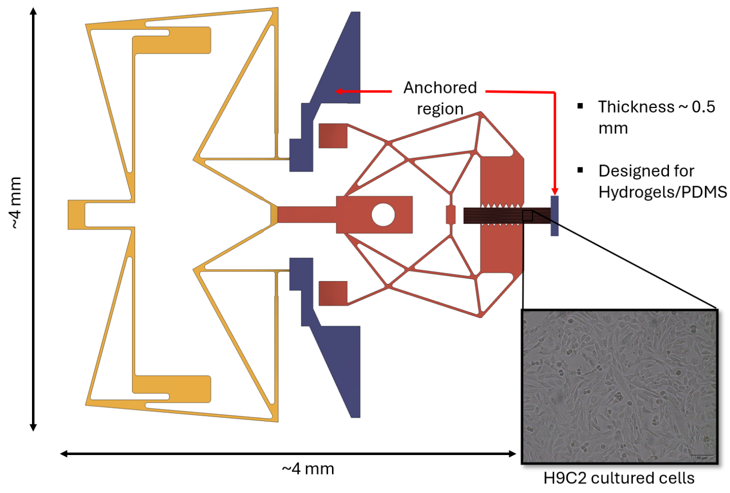
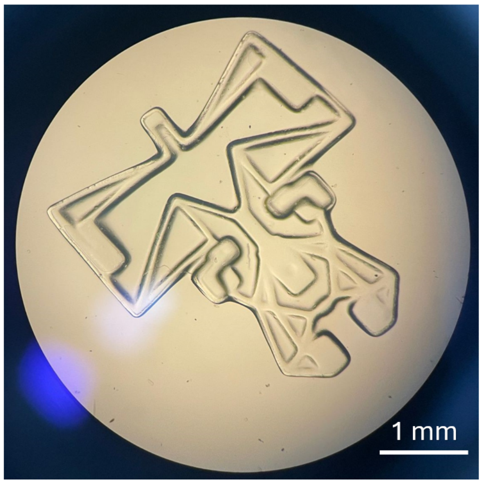
Thurston, RH (1895). "The Animal as a Machine and Prime Mover". In: *Science* 1.14.

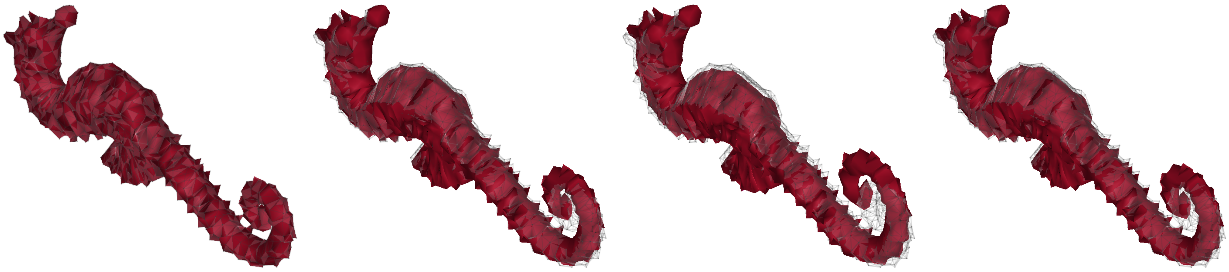
[1] https://mecheng.iisc.ac.in/m2d2/CMcollection/



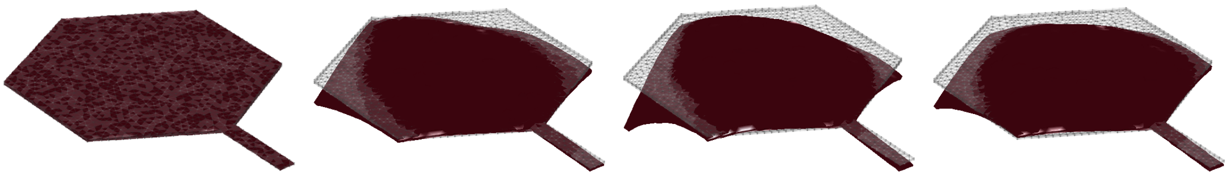
(a) Schematic of cell-actuated compliant mechanism



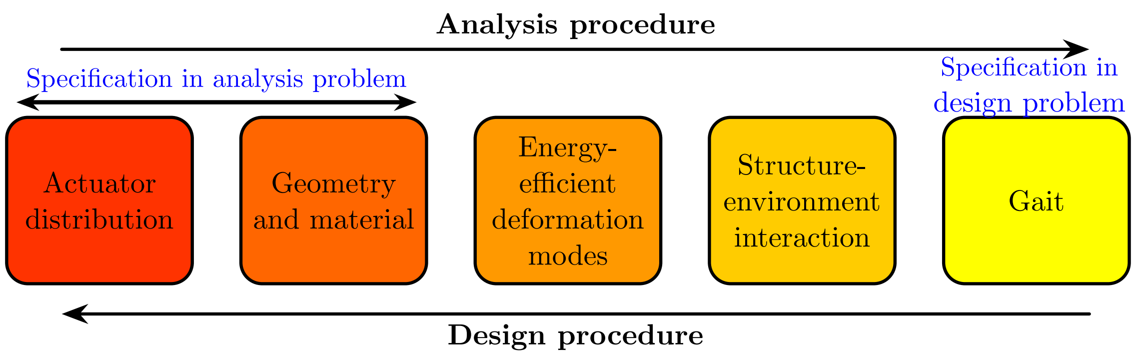
(b) Bioprinted mechanism (pilot test)



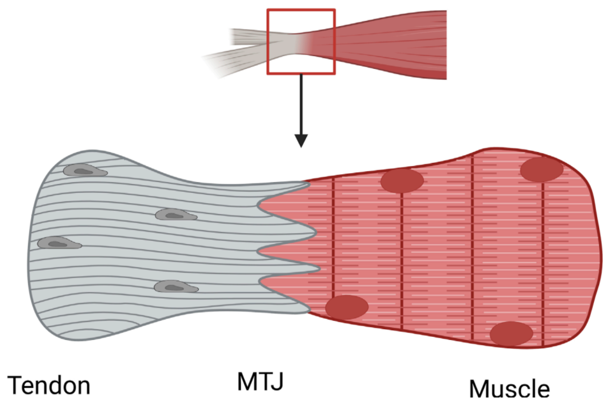
(c) $t = 0$ s (d) $t = 0.25$T (e) $t = 0.5$T (f) $t = 0.75$T



(g) $t = 0$ s (h) $t = 0.25$T (i) $t = 0.5$T (j) $t = 0.75$T



(k) Overview of objective 3



(l) Myotendinous interface found in animals (image from Adams et al. (2025))

Figure 2: Methodology and preliminary results